

## Problem Set 2

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| <b>Course Name:</b><br>Measurement and Control | <b>LP / Year:</b><br>1 / 2025      | <b>Course Code:</b><br>R0005E                  |
| <b>Number of questions:</b><br>7               | <b>Release Date:</b><br>2025-09-14 | <b>Submission Date:</b><br>2025-09-26 by 08:15 |

1. (a) Mention and explain the two main approaches to develop a model of a dynamical system.
- (b) Based on observation of the step response, most dynamical systems can be approximated as either a first order or a second order system. Specify the main difference between the first and second order system, based on its response. Please explain using graphical illustration to help.
- (c) Provide the main parameters of each first order and second order transfer functions. Provide an example of each transfer function, and determine the value of the corresponding parameters.

2. Given the following pairs of second-order system specifications:

- i.  $\%M_p = 10\%$ ,  $T_p = 5$  seconds.
- ii.  $T_s = 7$  seconds,  $T_p = 3$  seconds.

For each system:

- (a) Find the transfer function and the pole locations.
- (b) Find the damping ratio  $\zeta$  and the natural frequency  $\omega_n$ .
- (c) Plot the unit step response using MATLAB or Simulink, and verify the correctness of the plot you obtain by comparing with the given specifications.

3. Given a block diagram of a complex system as shown in Figure 1:

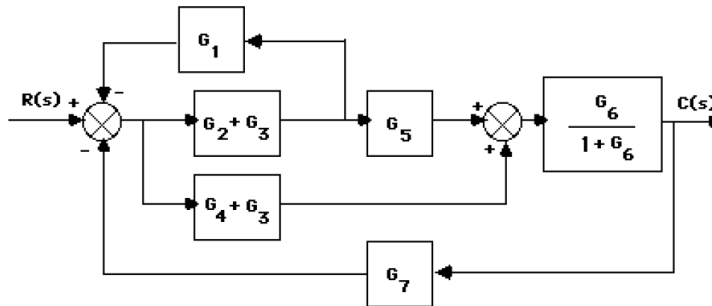


Figure 1: Block diagram of a complex system

- (a) Simplify the block diagram and find the transfer functions  $\frac{C(s)}{R(s)}$ .
- (b) If we consider  $G_7(s)$  as the feedback component of the closed-loop system, find the total feed-forward transfer function.

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4. A control system has a state-space model represented by:

$$\dot{x} = Ax + Bu = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u; \quad y = Cx + Du = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x$$

- (a) Find the eigenvalue of the state matrix  $A$ .
- (b) Show that the equivalent transfer function of the system is

$$\frac{Y(s)}{U(s)} = \frac{3s + 5}{s^2 + 4s + 6}$$

Hint: Use the formula  $\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$ , with  $s$  the Laplace variable.

- (c) Find the poles of the system and determine the stability of the system.
- (d) Compare these poles with the eigenvalues that you obtain in (a). Comment on the relation between the stability of the system in a state-space representation and in transfer function representation.
- (e) Use the MATLAB command `tf2ss` to obtain the state-space model of  $\frac{Y(s)}{U(s)}$  and compare the result to the original state-space model above.

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5. Consider the position control system in Figure 2, in which  $K$  is an adjustable gain parameter and

$$G(s) = \frac{6}{s^2 + 4.2s + 36}$$

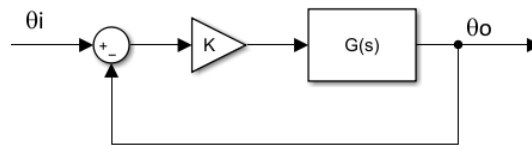


Figure 2: Block diagram of a position control system.

- (a) For  $K=1$ , find the closed-loop transfer function  $T(s)$ .
- (b) Find the steady state error of each the open-loop system  $G(s)$  and the closed-loop system  $T(s)$ , with respect to a unit step input ( $\Theta_i(s) = \frac{1}{s}$ ).
- (c) Explain your answer by relating the steady-state errors that you find, to the type of the system.
- (d) Compare the steady state errors that you obtained in (b), which one is larger and give explanation.

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6. Repeat Questions 4(a) and (b), for the transfer function:

$$G_1(s) = \frac{s + 6}{s^2 + 4.2s + 36}$$

- (a) Compare the steady state error of the closed-loop system  $T_1(s)$ , with the one you obtainefor  $T(s)$  in Question 4(b).
  - (b) Explain the effect of a zero in  $G_1(s)$  to the closed-loop system  $T_1(s)$ , taking the comparison with  $T(s)$ .
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7. The block diagram of a control system is depicted in Figure 3.

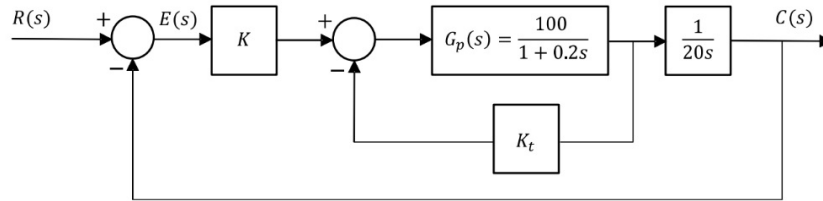


Figure 3: Block diagram of a double loop system

- (a) Find the closed-loop transfer function  $T(s) = \frac{C(s)}{R(s)}$ .
- (b) Determine the stability of the closed-loop system if  $K = 1$  and  $K_t = 0$ .
- (c) Find the steady-state error in unit-ramp response ( $r(t) = t$ ) in terms of  $K$  and  $K_t$ .
- (d) Discuss the effects of varying the values of  $K$  and  $K_t$  to the steady-state error.

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